

ME 333
MECHANICS OF MATERIALS

Lecture 27: Infinite Plate with a Punched Circular Hole

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Sem I, 2025-2026

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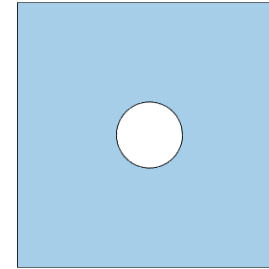
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Infinite Plate with a Punched Circular Hole

Plate with a circular hole under hydrostatic plane stress

Objective : Determine the radial and hoop stress fields in an infinite plate with a small circular hole embedded in it under equi-biaxial tension (hydrostatic plane stress).



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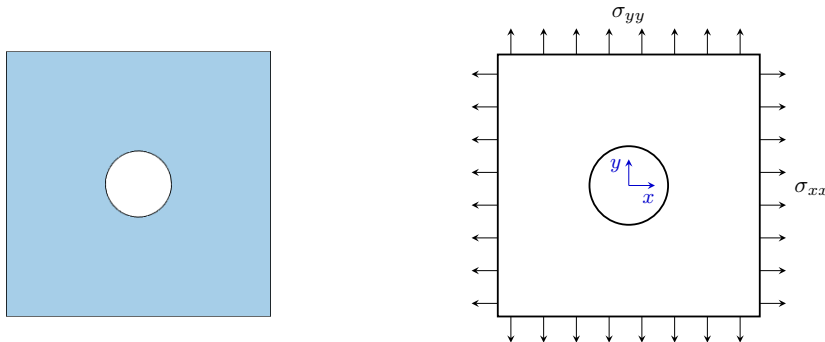
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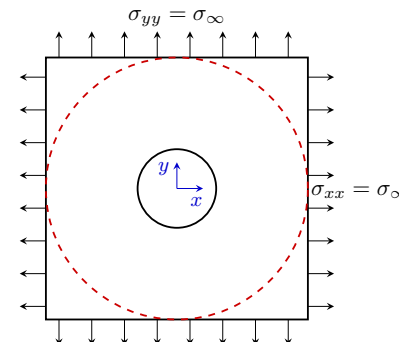
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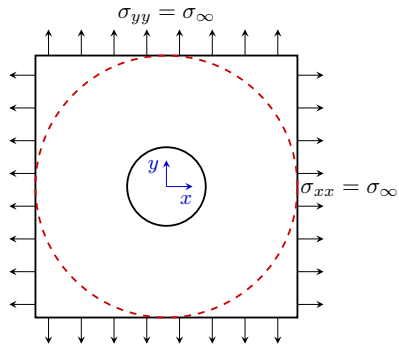
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• As $R_o \rightarrow \infty$:

$$\begin{aligned}\sigma_{rr} &= \frac{P_i R_i^2 - P_o R_o^2}{(R_o^2 - R_i^2)} + \frac{R_i^2 R_o^2 (P_o - P_i)}{r^2 (R_o^2 - R_i^2)} \\ &= \frac{R_o^2 (P_i (R_i^2 / R_o^2) - P_o)}{R_o^2 (1 - (R_i^2 / R_o^2))} + \frac{R_i^2 R_o^2 (P_o - P_i)}{r^2 R_o^2 (1 - (R_i^2 / R_o^2))} \\ &= -P_o + P_o \frac{R_i^2}{r^2} \\ &= \sigma_{\infty} - \frac{\sigma_{\infty} R^2}{r^2} = \sigma_{\infty} \left[1 - \left(\frac{R}{r} \right)^2 \right]\end{aligned}$$

Plate with a circular hole under hydrostatic plane stress



- The stress field description is :

$$\sigma_{rr} = \sigma_{\infty} - \frac{\sigma_{\infty} R^2}{r^2} = \sigma_{\infty} \left[1 - \left(\frac{R}{r} \right)^2 \right]$$

$$\sigma_{\theta\theta} = \sigma_{\infty} + \frac{\sigma_{\infty} R^2}{r^2} = \sigma_{\infty} \left[1 + \left(\frac{R}{r} \right)^2 \right]$$

- Stress concentration factor is defined as :

$$k = \frac{\sigma_{\theta\theta}}{\sigma_{\infty}} = \frac{\text{max stress}}{\text{nominal stress}}$$

- NOTE : Stress concentration factor does not depend on the absolute size or radius of the hole**

Plate with a circular hole under uniaxial stress

Objective : Determine the radial and hoop stress fields in an infinite plate with a small circular hole embedded in it under uniaxial tension.

- Boundary Conditions

$$@ x \rightarrow \infty : \quad \sigma_{xx} = \sigma_{\infty}; \quad \sigma_{yy} = \sigma_{xy} = 0$$

- Considering an arbitrary point far away from the hole, i.e., ($r \rightarrow \infty, \theta$) is:

$$\sigma_{rr} = \frac{\sigma_{xx} + \sigma_{yy}}{2} + \frac{\sigma_{xx} - \sigma_{yy}}{2} \cos 2\theta + \sigma_{xy} \sin 2\theta$$

$$\sigma_{\theta\theta} = \frac{\sigma_{xx} + \sigma_{yy}}{2} - \frac{\sigma_{xx} - \sigma_{yy}}{2} \cos 2\theta + \sigma_{xy} \sin 2\theta$$

$$\sigma_{r\theta} = -\frac{\sigma_{xx} - \sigma_{yy}}{2} \sin 2\theta + \sigma_{xy} \cos 2\theta$$

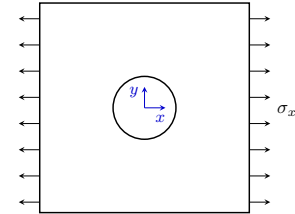


Plate with a circular hole under uniaxial stress

Objective : Determine the radial and hoop stress fields in an infinite plate with a small circular hole embedded in it under uniaxial tension.

- Considering an arbitrary point far away from the hole, i.e., ($r \rightarrow \infty, \theta$) is:

$$\sigma_{rr} = \frac{\sigma_{\infty}}{2} + \frac{\sigma_{\infty}}{2} \cos 2\theta$$

$$\sigma_{\theta\theta} = \frac{\sigma_{\infty}}{2} - \frac{\sigma_{\infty}}{2} \cos 2\theta$$

$$\sigma_{r\theta} = -\frac{\sigma_{\infty}}{2} \sin 2\theta$$

- Boundary Conditions:

$$@ (r = R_i, \theta) \quad \sigma_{rr} = \sigma_{r\theta} = 0$$

Note: No condition on tangential and hoop stresses.

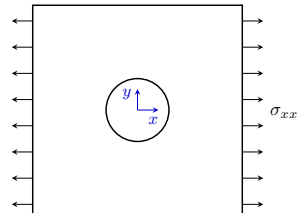


Plate with a circular hole under uniaxial stress

Objective : Determine the radial and hoop stress fields in an infinite plate with a small circular hole embedded in it under uniaxial tension.

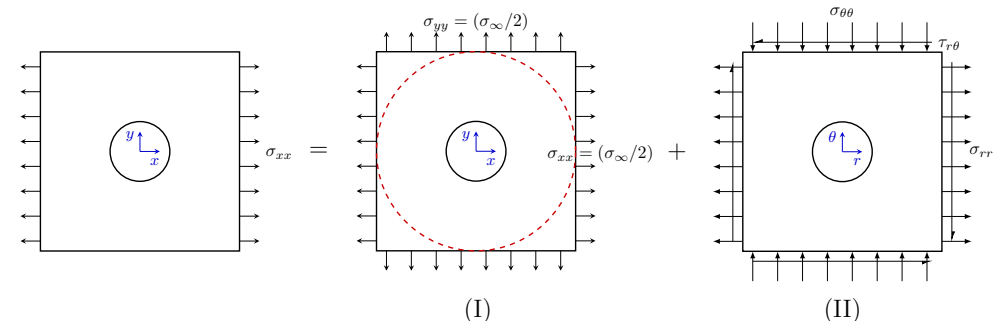


Plate with a circular hole under uniaxial stress

- For (I) the stress field description $\forall(r, \theta)$ is:

$$\sigma_{rr} = \frac{\sigma_{\infty}}{2} \left[1 - \frac{R_i^2}{r^2} \right]; \quad \sigma_{\theta\theta} = \frac{\sigma_{\infty}}{2} \left[1 + \frac{R_i^2}{r^2} \right]; \quad \sigma_{r\theta} = 0$$

- For (II) to obtain the stress field description $\forall(r, \theta)$ we need to solve the general bi-harmonic equation, i.e.,

$$\nabla^4 \phi = \frac{\partial^4 \phi}{\partial r^4} + \frac{2}{r^2} \frac{\partial^4 \phi}{\partial r^2 \partial \theta^2} + \frac{1}{r^4} \frac{\partial^4 \phi}{\partial \theta^4} + \frac{2}{r} \frac{\partial^3 \phi}{\partial r^3} + \frac{2}{r^3} \frac{\partial^3 \phi}{\partial r \partial \theta^2} - \frac{1}{r^2} \frac{\partial^2 \phi}{\partial r^2} + \frac{4}{r^4} \frac{\partial^2 \phi}{\partial \theta^2} + \frac{1}{r^3} \frac{\partial \phi}{\partial r}$$

Considering a solution of the form $\phi(r, \theta) = f(r) \cos 2\theta$ noting the periodicity in θ , the solution to the above equation results in:

$$\phi(r, \theta) = \left(C_1 + C_2 r^2 + C_3 r^4 + \frac{C_4}{r^2} \right) \cos 2\theta$$

Challenge : Determine the unknown coefficients.

Plate with a circular hole under uniaxial stress

- The stress field can then be written as :

$$\sigma_{rr} = \frac{1}{r} \frac{\partial \phi}{\partial r} + \frac{1}{r} \frac{\partial^2 \phi}{\partial \theta^2} = - \left(2C_2 + 4 \frac{C_1}{r^2} + 6 \frac{C_4}{r^4} \right) \cos 2\theta$$

$$\sigma_{\theta\theta} = \frac{\partial^2 \phi}{\partial r^2} = \left(2C_2 + 12C_3 r^2 + 6 \frac{C_4}{r^4} \right) \cos 2\theta$$

$$\sigma_{r\theta} = \frac{1}{r^2} \frac{\partial \phi}{\partial \theta} - \frac{1}{r} \frac{\partial^2 \phi}{\partial r \partial \theta} = \left(2C_2 + 6C_3 r^2 - 2 \frac{C_1}{r^2} - 2 \frac{C_4}{r^4} \right) \sin 2\theta$$

- Boundary conditions :

$$@ (R_i, \theta) : \sigma_{rr} = \sigma_{r\theta} = 0; \quad @ (r \rightarrow \infty, \theta) : \sigma_{r\theta} = -\frac{\sigma_{\infty}}{2} \sin 2\theta$$

Applying the boundary conditions will result in:

$$C_1 = \frac{\sigma_{\infty}}{2} R_i^2; \quad C_2 = -\frac{\sigma_{\infty}}{4}; \quad C_3 = 0; \quad C_4 = -\frac{\sigma_{\infty}}{4} R_i^4$$

Plate with a circular hole under uniaxial stress

- The stress field in (II) $\forall(r, \theta)$ can be written as :

$$\sigma_{rr} = \frac{\sigma_{\infty}}{2} \left(1 - 4 \frac{R_i^2}{r^2} + 3 \frac{R_i^4}{r^4} \right) \cos 2\theta$$

$$\sigma_{\theta\theta} = -\frac{\sigma_{\infty}}{2} \left(1 + 3 \frac{R_i^4}{r^4} \right) \cos 2\theta$$

$$\sigma_{r\theta} = \frac{\sigma_{\infty}}{2} \left(-1 - 2 \frac{R_i^2}{r^2} + 3 \frac{R_i^4}{r^4} \right) \sin 2\theta$$

- The stress field description for a plate with a circular hole subjected to uniaxial stress can be obtained by superposing the stresses determined for cases (I) and (II).

Plate with a circular hole under uniaxial stress

- The net stress field under uniaxial loading $\forall(r, \theta)$ can be written as :

$$\sigma_{rr} = \frac{\sigma_{\infty}}{2} \left[1 - \frac{R_i^2}{r^2} \right] + \frac{\sigma_{\infty}}{2} \left(1 - 4 \frac{R_i^2}{r^2} + 3 \frac{R_i^4}{r^4} \right) \cos 2\theta$$

$$\sigma_{\theta\theta} = \frac{\sigma_{\infty}}{2} \left[1 + \frac{R_i^2}{r^2} \right] - \frac{\sigma_{\infty}}{2} \left(1 + 3 \frac{R_i^4}{r^4} \right) \cos 2\theta$$

$$\sigma_{r\theta} = \frac{\sigma_{\infty}}{2} \left(-1 - 2 \frac{R_i^2}{r^2} + 3 \frac{R_i^4}{r^4} \right) \sin 2\theta$$

- When $r = a$ and $\theta = (\pi/2)$ it results in a maximum value of the hoop stress $\sigma_{\theta\theta} = 3\sigma_0$. Here, 3 is the stress concentration factor.
- Similar analysis of a plate with a circular hole subjected to pure shear will result in a stress concentration factor of 4.

Plate with a circular hole under uniaxial stress

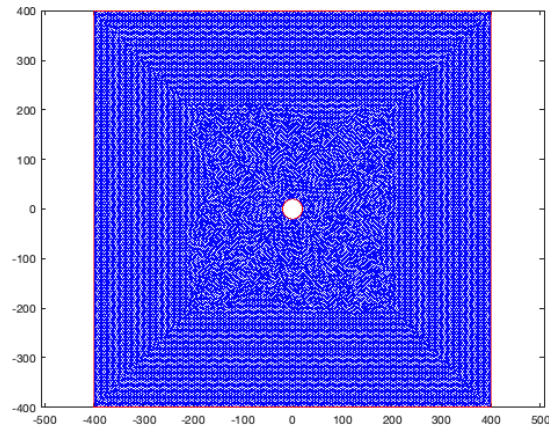


Plate with a circular hole under uniaxial stress

Normal stress along x-direction

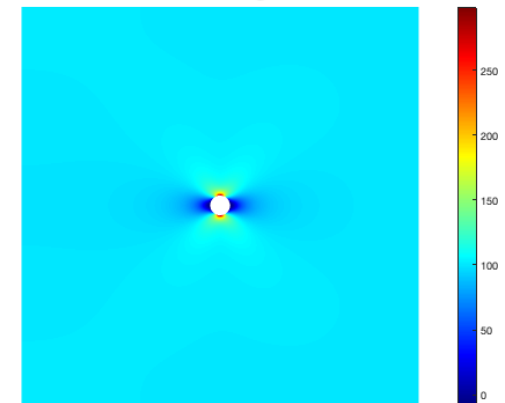


Plate with a circular hole under uniaxial stress

Normal stress along y-direction

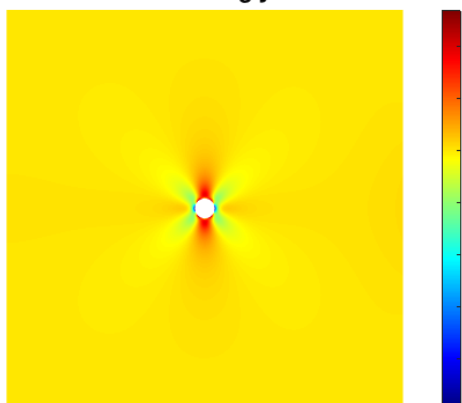


Plate with a circular hole under uniaxial stress

Shear stress in x-y plane

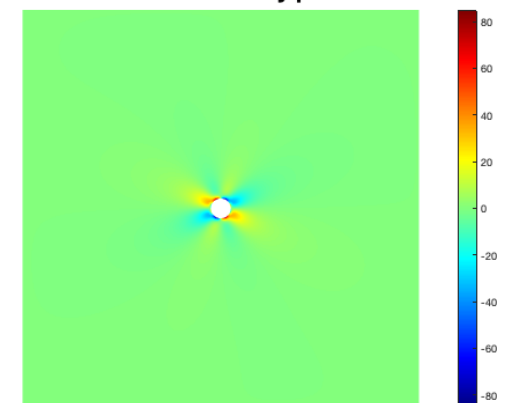


Plate with a circular hole in a strip under uniaxial stress

Normal stress along x-direction
 $d=20, H=100, \sigma_x = 100$

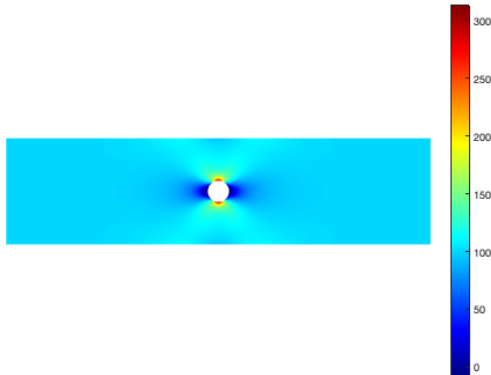


Plate with a circular hole in a strip under uniaxial stress

Normal stress along x-direction
 $d=20, H=50, \sigma_x = 100$

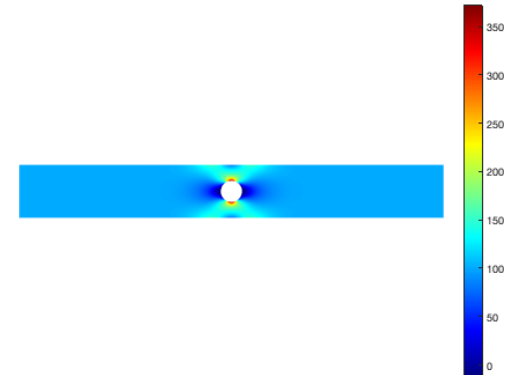


Plate with a circular hole near edge under uniaxial stress

Normal stress along x-direction
 $d=20, H,W=400, \sigma_x = 100$

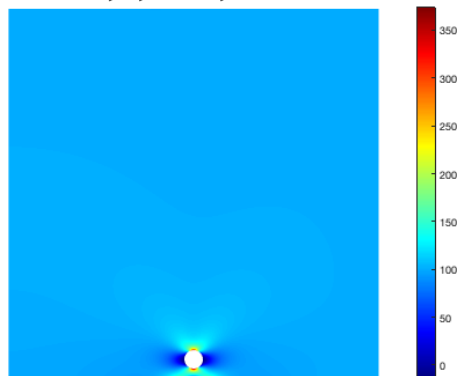


Plate with a circular hole near edge under uniaxial stress

Normal stress along x-direction
 $d=20, H,W=400, \sigma_x = 100$

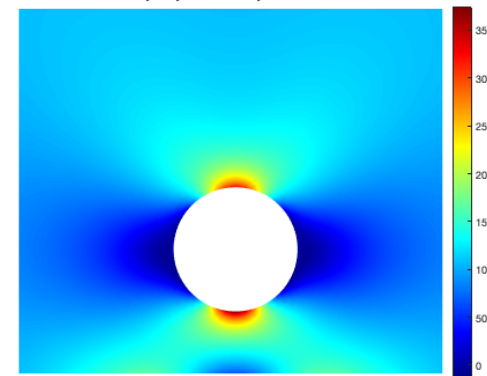
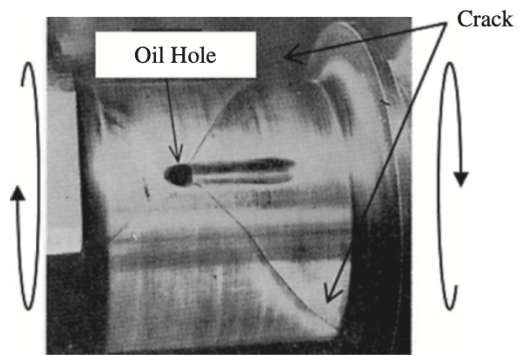


Plate with a circular hole near edge under uniaxial stress



Working Assignment

Work out all steps in the above problem